

Yibo Chen

Undergraduate Researcher in Artificial Intelligence

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Research Interests

AI agents and model behavior; reasoning-to-action gaps; revision and behavioral internalization; procedural control and decision authority; evaluation methodology; cognitive architectures for learned control, memory, and adaptation.

Education

Tsinghua University - Department of Automation, Undergraduate 2025-Present

Research Experience

Independent Research - Tsinghua University 2026-Present

- Lead end-to-end studies of language-model agents using controlled, paired, and stage-resolved experiments; separate process behavior from endpoint success and test robustness across tasks, models, and operationalizations.
- Produced four single-author public preprints on coding-agent trajectories, implementation fidelity, procedural objectives, and whether model judgements become causally binding for later behavior.

Collaborative Research - Beijing Institute for General Artificial Intelligence (BIGAI) Summer 2026-Present

- Contribute to research on language-model reasoning and representation, including data review, experimental redesign, probing, quantitative analysis, and circuit-discovery work.

Preprints

Behavioral Success Without Internalization: When Scientific Judgements Fail to Bind Language-Model Policies 2026
TechRxiv preprint

When Evidence Becomes a Veto: Procedural Objectives and Decision-Authority Inflation in LLM Agents 2026
TechRxiv preprint

What Changes, When? Intervention Effects Along Coding-Agent Trajectories 2026
TechRxiv preprint

Same Contract, Different Product Role: A Controlled Study of Coding-Agent Implementation Fidelity 2026
TechRxiv preprint

All four preprints are single-author works.

Methods & Technical Skills

Controlled and paired experimental design; stage-resolved and causal evaluation; bootstrap confidence intervals and sensitivity analysis; LLM/agent evaluation pipelines; Python; PyTorch; Hugging Face Transformers; Linux/SSH; Git; LaTeX.